

Jeevan Kumar Banoth - 02105145
CIS 602-7103: Special Topics in CIS

Homework – 4:

Task 1: Reviewing the role of IAM in accessing and configuring Amazon Redshift

To open the IAM Console, do the following: Navigate to the AWS Management Console. Type “IAM” into the search box and select it.

Find the role which name is MyRedshiftRole: Navigate to Roles by selecting it on the left-hand side of the IAM console. Enter “MyRedshiftRole” into the search box. Click on the link for MyRedshiftRole as soon as you see it.

The screenshot displays the AWS IAM console interface. The top navigation bar includes the AWS logo, a search bar, and the user's account information. The left-hand sidebar shows the 'Identity and Access Management (IAM)' section with a search bar and a list of navigation options: Dashboard, Access management (User groups, Users, Roles, Policies, Identity providers, Account settings), Access reports (Access Analyzer, External access, Unused access, Analyzer settings, Credential report, Organization activity), CloudShell, and Feedback.

The main content area is titled 'IAM Dashboard' and contains several sections:

- IAM resources:** A table showing the count of resources in the AWS Account.
- What's new:** A section with updates for features in IAM.
- AWS Account:** A section showing account details like Account ID, Account Alias, and Sign-in URL.
- Tools:** A section with a link to the Policy simulator.

The 'IAM resources' table is as follows:

User groups	Users	Roles	Policies	Identity providers
0	0	19	4	0

The 'What's new' section lists updates for features in IAM, including AWS IAM Access Analyzer, AWS IAM Access Analyzer recommendations, AWS Launches Console-based Bulk Policy Migration, and IAM Roles Anywhere.

The 'AWS Account' section shows the Account ID (742204132534), Account Alias, and Sign-in URL for IAM users in this account.

The 'Tools' section includes a link to the Policy simulator, which evaluates the policies that you choose and determines the effective permissions for each of the actions that you specify.

The bottom section of the screenshot shows the details of the 'myRedshiftRole'.

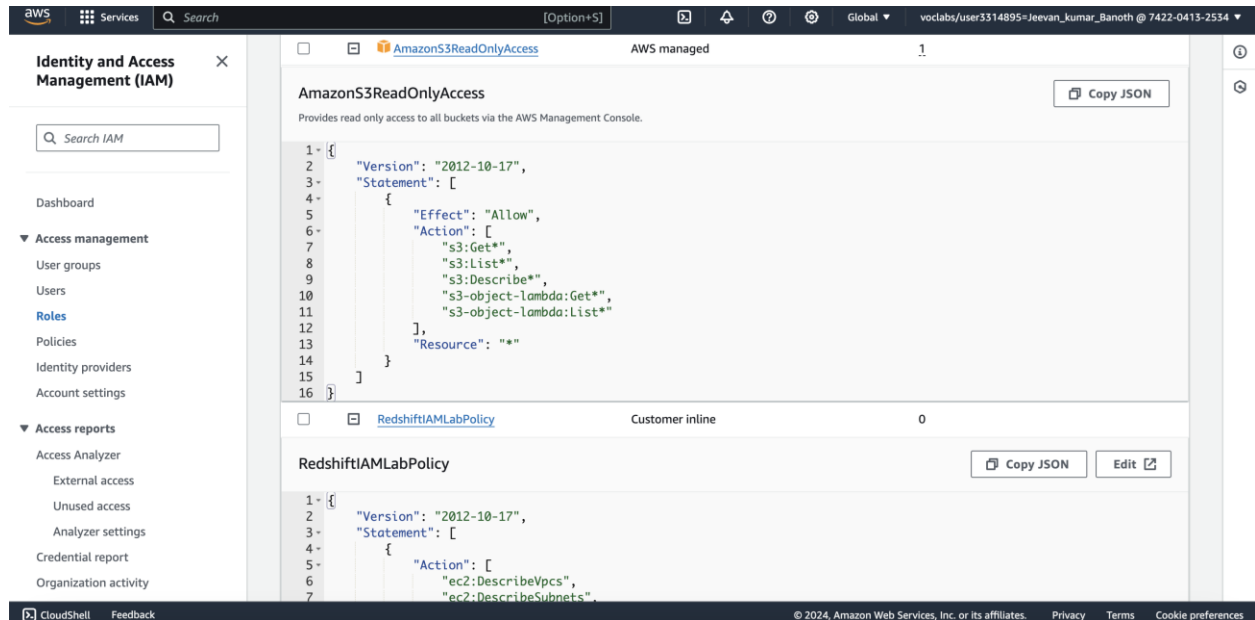
The 'myRedshiftRole' page includes a 'Summary' section with the following information:

- Creation date: June 12, 2024, 21:33 (UTC-04:00)
- Last activity: -
- ARN: `arn:aws:iam::742204132534:role/myRedshiftRole` (copied)
- Maximum session duration: 1 hour

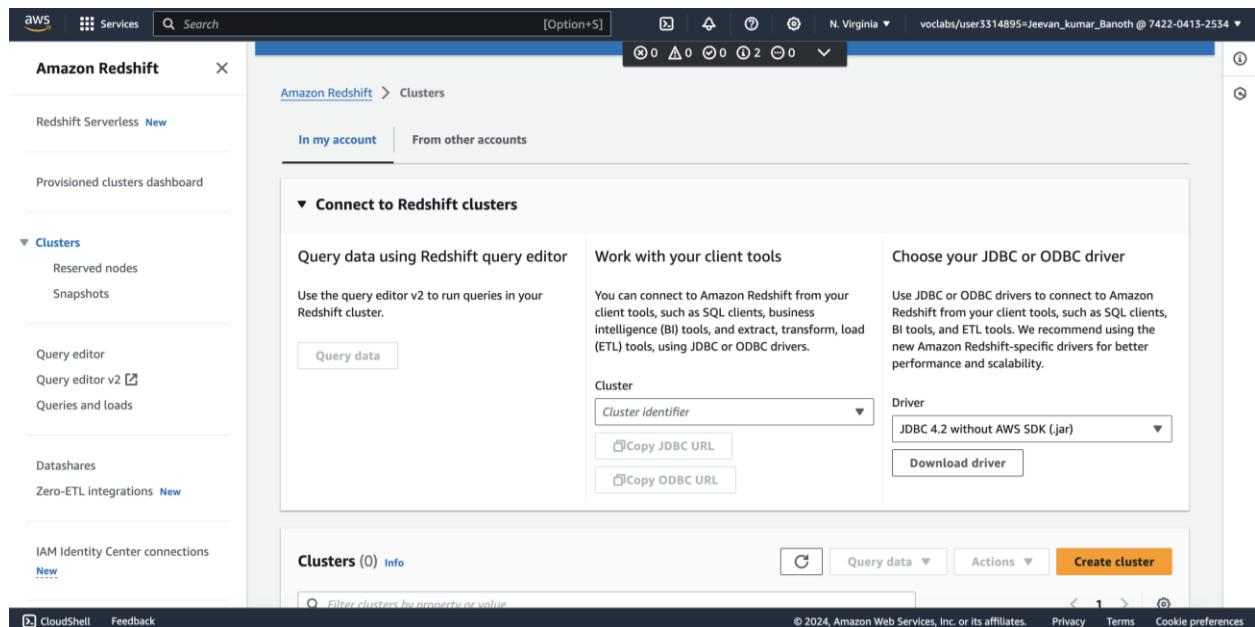
The 'Permissions' section shows the role has 2 permissions policies attached. The table below lists the attached policies:

Policy name	Type	Attached entities
AmazonS3ReadOnlyAccess	AWS managed	1

- In the **Summary** section at the top of the page, find the **ARN** for the role. I have copied the ARN for later use and pasted it in notepad.



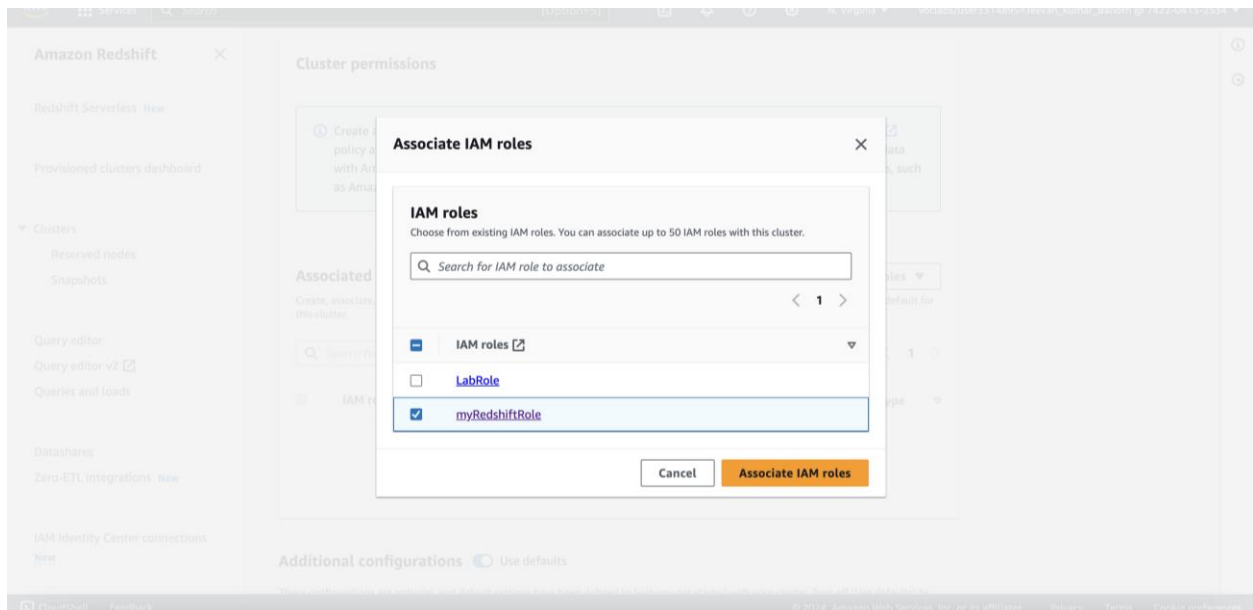
As you can see above, we will be viewing the policies of AmazonS3ReadOnlyAccess and RedshiftIAMLab policy.



Next, we will be opening Amazon Redshift from the search box and then we will be moving towards clusters and start creating new clusters.

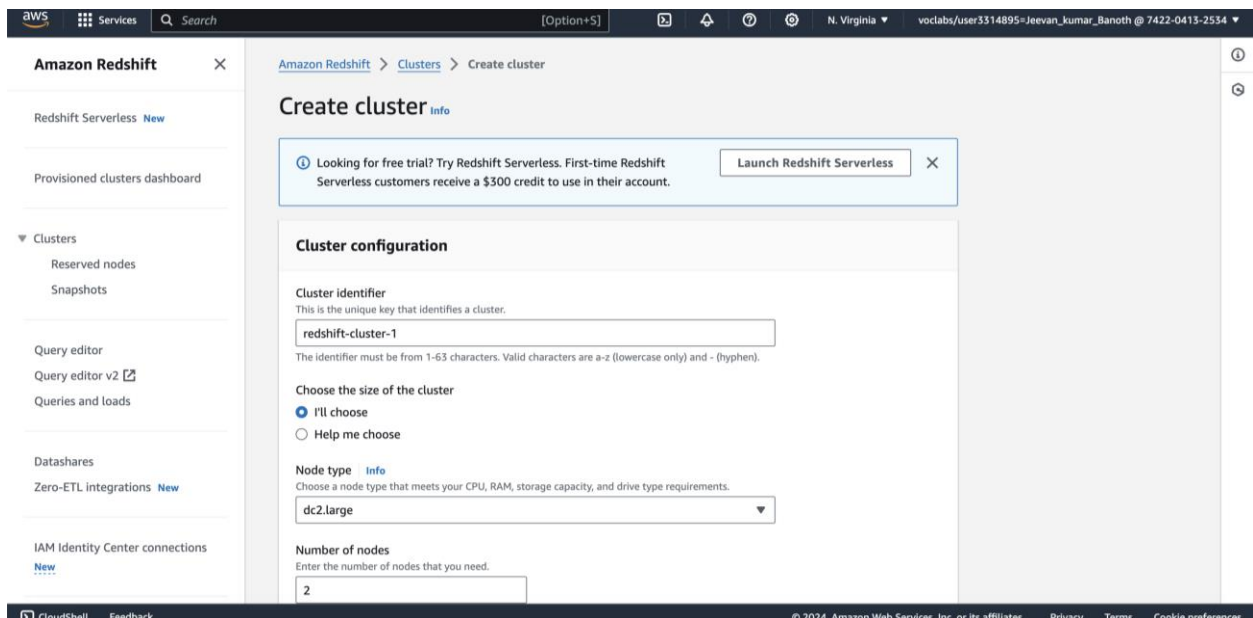
You should scroll down to the Permissions part. Try to find and open the AmazonS3ReadOnlyAccess policy. Please check if Amazon Redshift has been granted

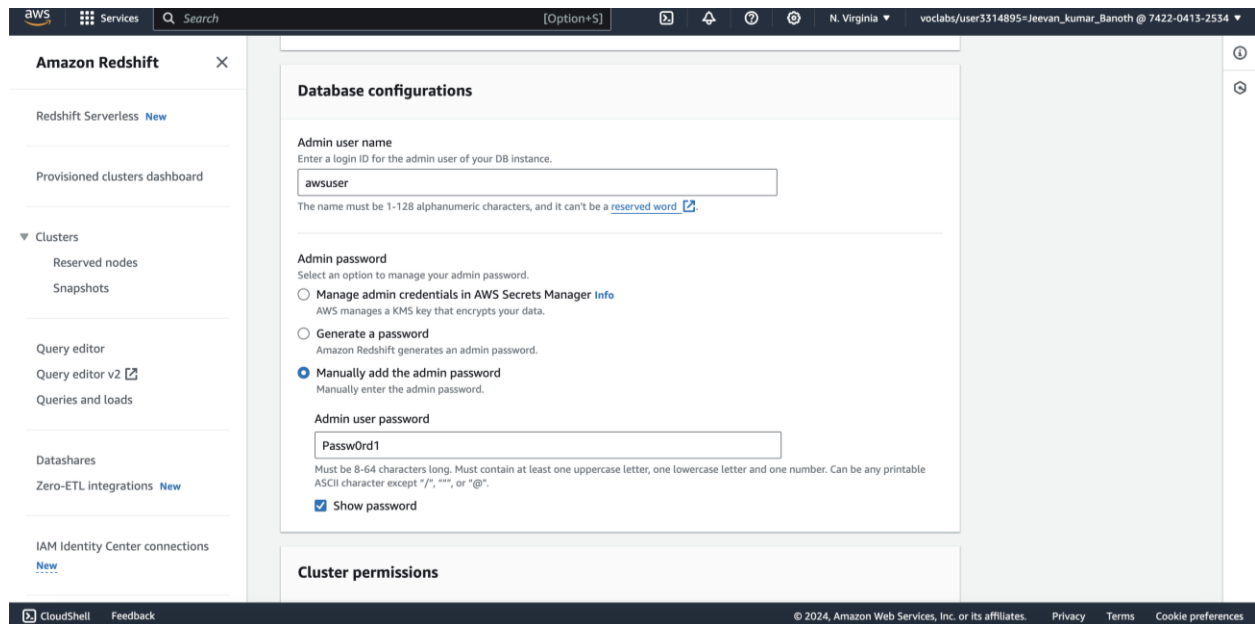
permissions to access and read data from S3 buckets. Check out the RedshiftIAMLabPolicy policy. Check if Amazon Redshift has permissions to describe Amazon EC2 and VPC resources.



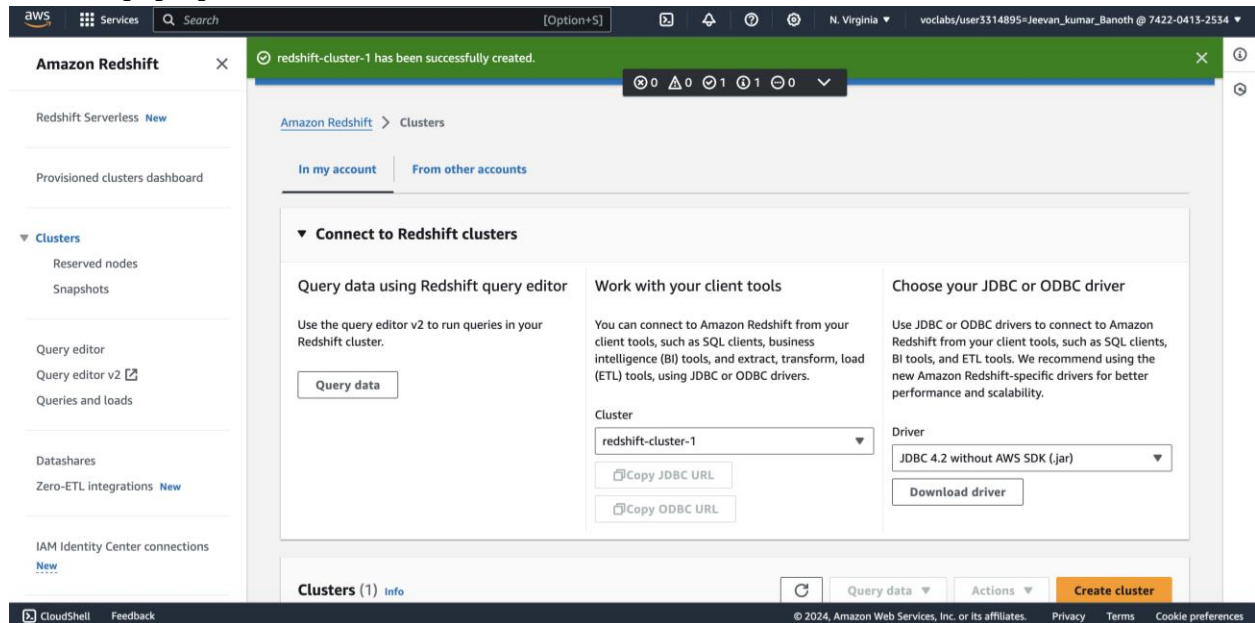
Task 2: Creating and Configuring a Redshift Cluster

"Navigate to Cluster using the menu on the left of Redshift console. Pick Create cluster. for Cluster Configuration: Cluster identifier: redshift-cluster-1 Production: Select Node type: dc2.large Number of nodes: 2", Use the database configurations as given in the manual.



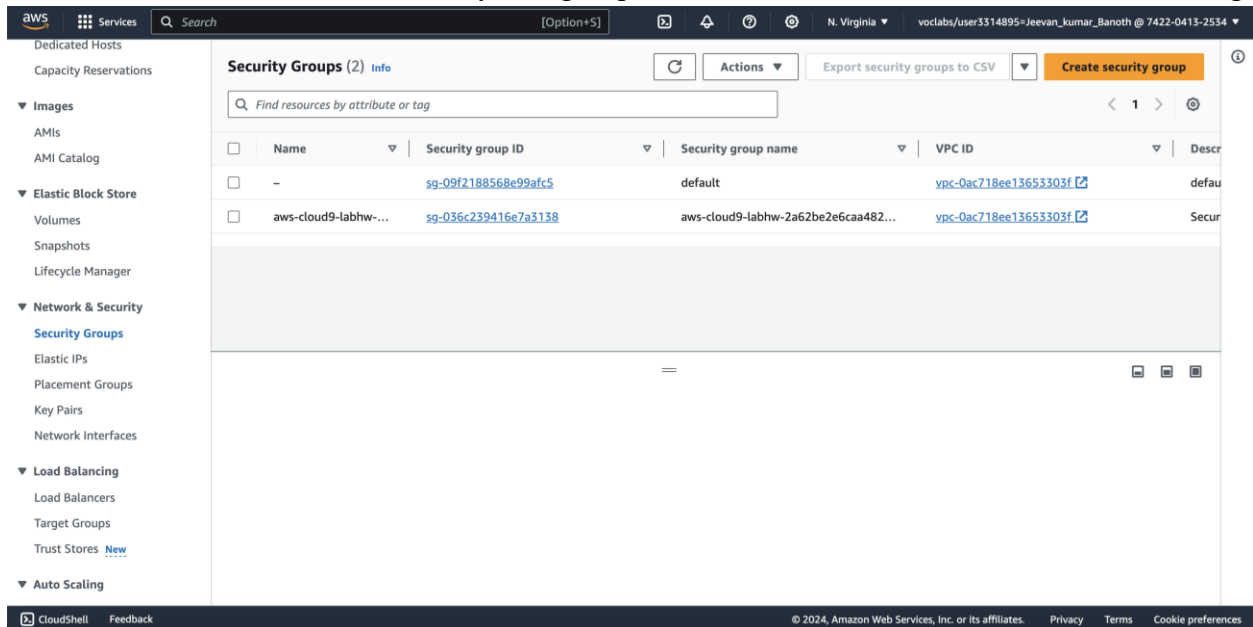


Select the Manage IAM roles option and associate it with IAM roles. Select MyRedshiftRole in the pop-up window, then click on Associate IAM roles.

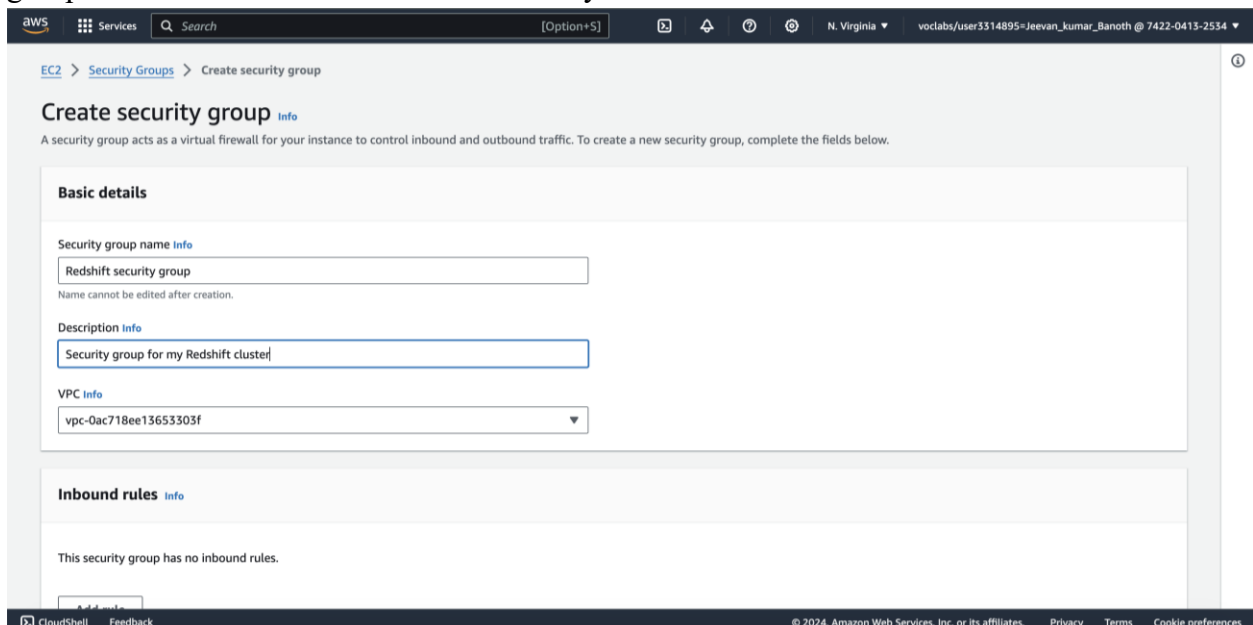


Ensure MyRedshiftRole is selected and there is (1/1) next to Associated IAM roles.

To do this, go to the tab called Security Groups in the section entitled Network & Security. Just click on “Create security group” and fill in the following:



Name your new security group “Redshift security group”. Give it a brief description e.g. “Security group for my Redshift cluster”.



In the Inbound Rules category: For that, we need to click “Add rule”. This new rule will be different from others since its type is Redshift and source can be anywhere on the Internet having IPv4 connectivity (‘Anywhere-IPv4’ 0.0.0.0/0). Let us now have this inbound rule specifying Redshift services only. At last, click on “Create security group”.

VPC Info

vpc-0ac718ee13653303f

Inbound rules

Type	Protocol	Port range	Source	Description - optional
Redshift	TCP	5439	Anyw... 0.0.0.0/0	Redshift inbound rule

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules

Type	Protocol	Port range	Destination	Description - optional
All traffic	All	All	Custom 0.0.0.0/0	

Ensure the cluster status is Available and return to the Amazon Redshift console after Selecting the redshift-cluster-1 cluster.

Provisioned clusters dashboard

Try Amazon Redshift Serverless | Purchase reserved nodes | **Create cluster**

Resources overview
Resource data for US East (N. Virginia) Region.

Total nodes	On-demand nodes	Reserved nodes	Reserved nodes available	Automated snapshots	Manual snapshots
2	2	0	0 (0 of 0 used)	3	0

Cluster overview (1)

Cluster	Status
redshift-cluster-1	Available

View all clusters

Cluster metrics

Any clusters | Last hour | View in CloudWatch

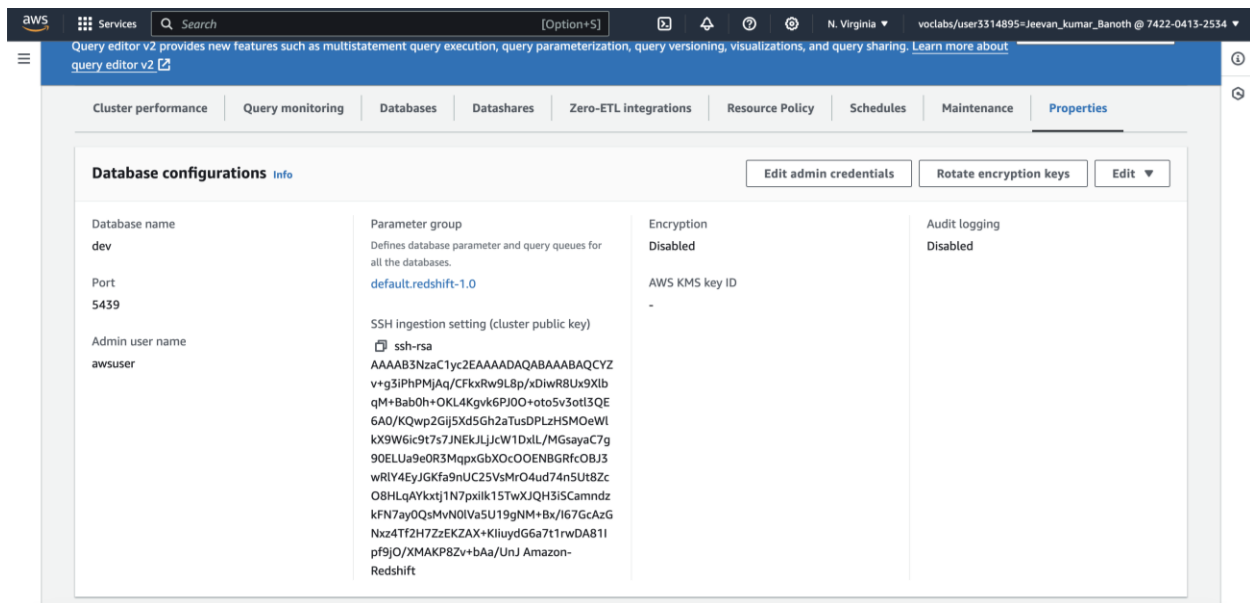
Datashares
Authorize other AWS accounts to access datashares created in this AWS account. Associate or decline datashares from other AWS accounts.

Require authorization

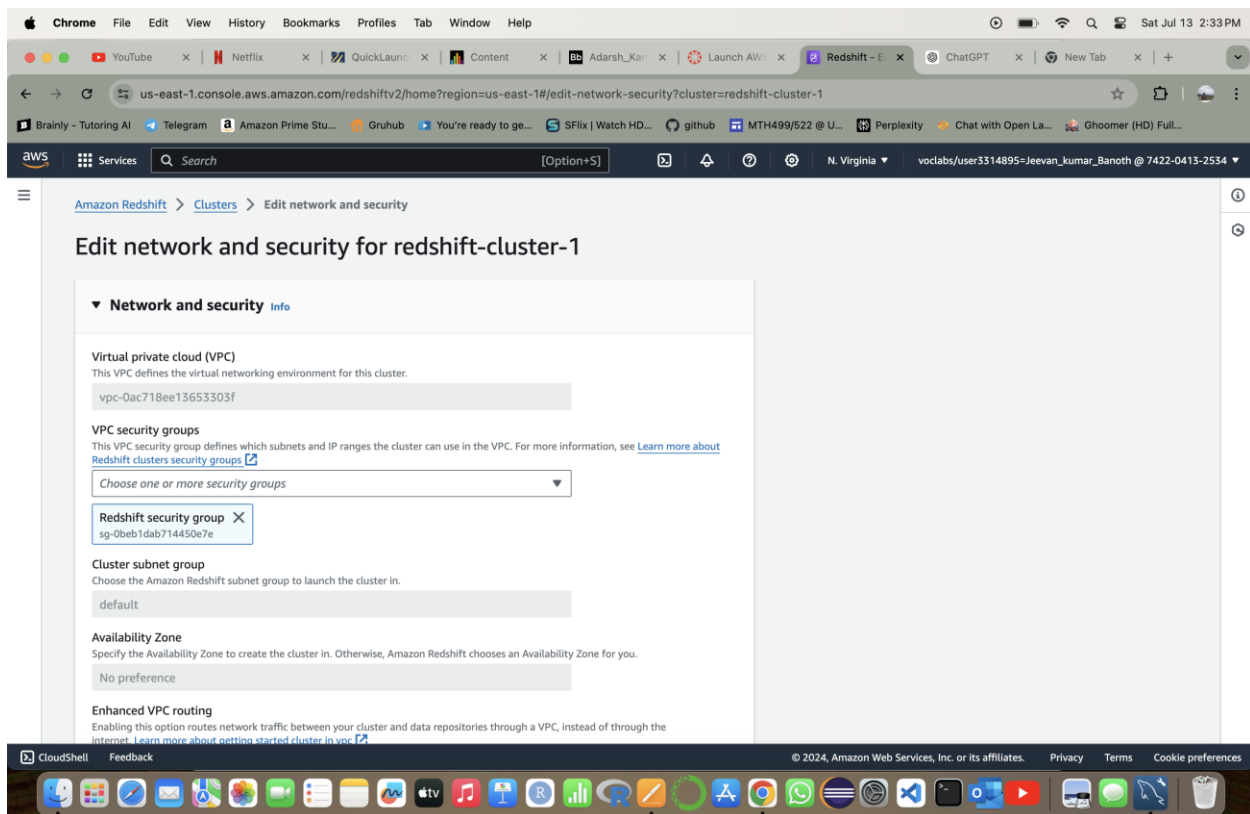
0

Require association

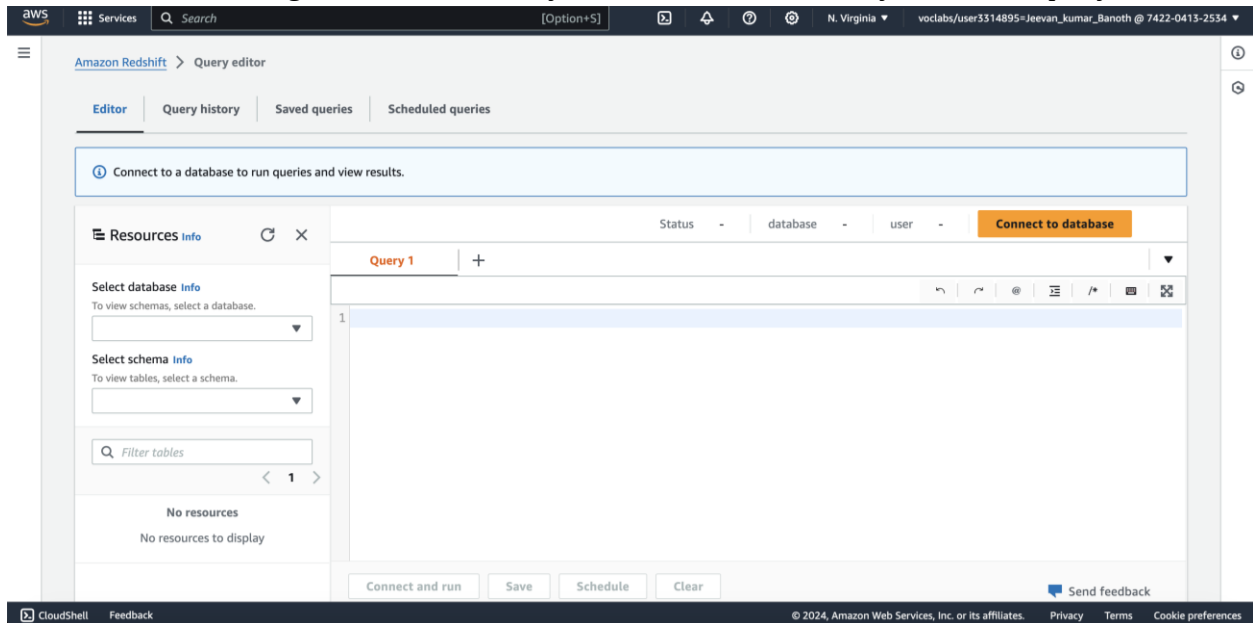
Enter the Properties tab. Go to the edits for Network. and security settings section. Select only Redshift security group and clear default For VPC security groups. Click to save changes. Wait until the status changes from Changing back to Available.



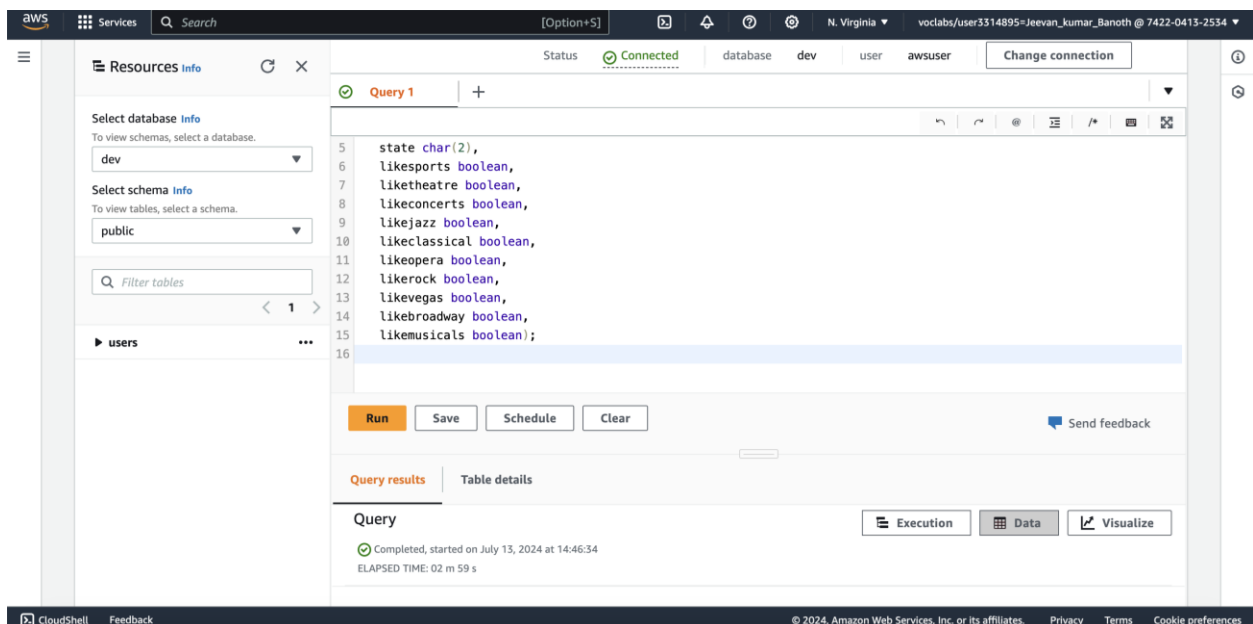
After opening Amazon Redshift, move to clusters and from that edit the Network and security. As shown below, chose the name as Redshift security group for Vpc security groups and remove the default one.



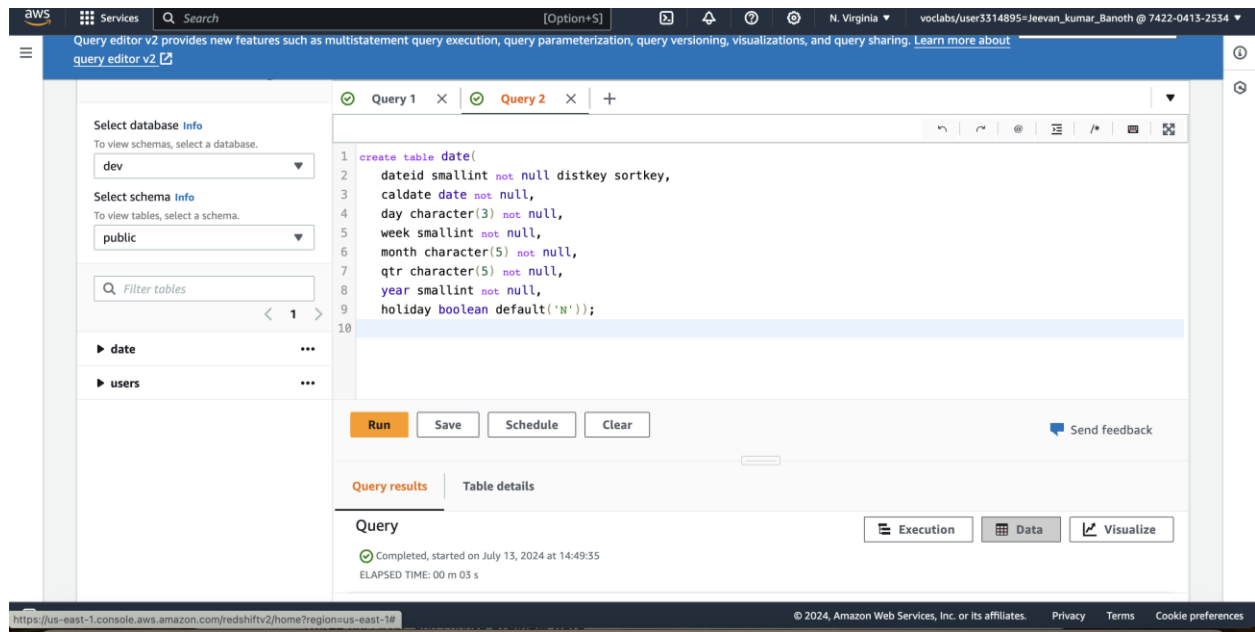
In the navigation pane, you should click on Clusters. After you identify redshift-cluster-1 cluster, select it. Navigate to Query data > Query in query editor.



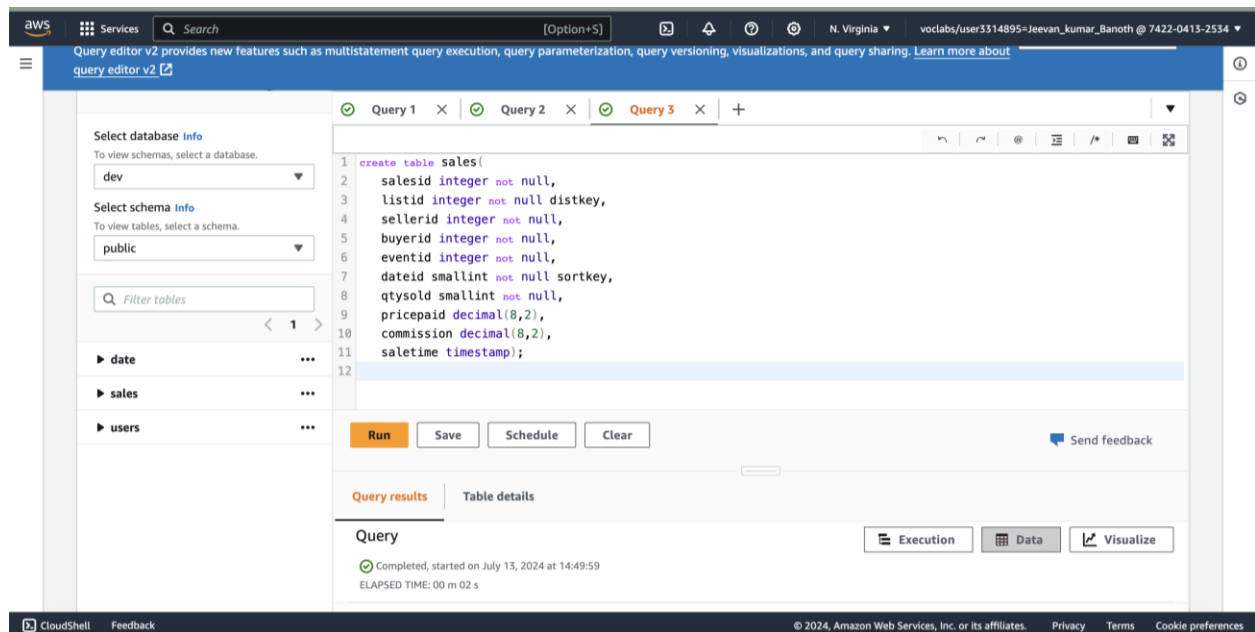
Choose Connect to database and set up the following: Connection: Create a new connection. Authentication: Secure credentials. Database name: dev. Database user: awsuser."



Run all given commands in the query boxes and check for the output and its tables.



So, as shown above, the first query will result in the table called data and the second query will result in a table called users as shown in the left side of above image.



The above query will result in the sales table as shown left in above picture.

Task 4: Loading Data from Amazon S3

The screenshot shows the AWS CloudShell interface with a SQL query editor. The query is as follows:

```
1 copy sales from 's3://aws-tc-largeobjects/CUR-TF-200-ACDSCI-1/Lab4/sales_tab.txt'
2 credentials 'aws_iam_role=arn:aws:iam::742204132534:role/myRedshiftRole'
3 delimiter '\t' timeformat 'MM/DD/YYYY HH:MI:SS' region 'us-west-2';
4
```

The interface includes a sidebar with 'Resources Info' showing the selected database 'dev' and schema 'public'. The main area displays the query editor with 'Query 4' selected. Below the editor, there are buttons for 'Run', 'Save', 'Schedule', and 'Clear'. The 'Query results' section shows that the query was completed successfully on July 13, 2024, at 14:56:35, with an elapsed time of 00 m 08 s.

Now, as shown above, we will be copying and loading the data into amazon S3 bucket.

The above command used for copying the sales table which has resulted above from here to S3.

The screenshot shows the AWS CloudShell interface with a SQL query editor. The query is as follows:

```
1 copy users from 's3://aws-tc-largeobjects/CUR-TF-200-ACDSCI-1/Lab4/allusers_tab.txt'
2 credentials 'aws_iam_role=arn:aws:iam::742204132534:role/myRedshiftRole'
3 delimiter '\t' region 'us-west-2';
4
```

The interface includes a sidebar with 'Resources Info' showing the selected database 'dev' and schema 'public'. The main area displays the query editor with 'Query 4' selected. Below the editor, there are buttons for 'Run', 'Save', 'Schedule', and 'Clear'. The 'Query results' section shows that the query was completed successfully on July 13, 2024, at 14:57:41, with an elapsed time of 00 m 08 s.

In that same manner we will be doing for the rest of it and in the place of ARN we will be replacing the one which have copied at the start of the lab.

The screenshot shows the AWS Redshift console interface. The top navigation bar includes the AWS logo, 'Services', a search bar, and the user's profile information. The main content area displays the 'Query results' tab for the 'sales' table. The table has columns: salesid, listid, sellerid, buyerid, eventid, dateid, qtytsold, pricepaid, and con. The data is presented in a table with 10 rows.

salesid	listid	sellerid	buyerid	eventid	dateid	qtytsold	pricepaid	con
150314	173969	48680	816	8762	1827	2	688.00	103
157751	206999	3003	157	6605	1827	1	1730.00	259
8325	8942	23600	1078	2557	1828	5	525.00	78.7
40331	45102	13005	1091	1756	1828	2	58.00	8.70
46807	52711	34388	1047	2046	1828	2	482.00	72.3
68413	77733	7199	1169	1611	1828	2	158.00	23.7
84206	96065	45532	1417	5607	1828	1	433.00	64.5
96274	109925	22409	194	1764	1828	1	104.00	15.6

This image is the preview of the results of the sales table which is generated.

The screenshot shows the AWS Redshift console interface. The top navigation bar includes the AWS logo, 'Services', a search bar, and the user's profile information. The main content area displays the 'Query results' tab for the 'date' table. The table has columns: dateid, caldate, day, week, month, qtr, year, and holiday. The data is presented in a table with 10 rows.

dateid	caldate	day	week	month	qtr	year	holiday
1828	2008-01-02	TH	1	JAN	1	2008	false
1829	2008-01-03	FR	1	JAN	1	2008	false
1830	2008-01-04	SA	2	JAN	1	2008	false
1833	2008-01-07	TU	2	JAN	1	2008	false
1839	2008-01-13	MO	3	JAN	1	2008	false
1841	2008-01-15	WE	3	JAN	1	2008	false
1848	2008-01-22	WE	4	JAN	1	2008	false
1855	2008-01-29	WE	5	JAN	1	2008	false
1861	2008-02-04	TU	6	FEB	1	2008	false
1866	2008-02-09	SU	7	FEB	1	2008	false

The above image is the preview of the date table which was generated earlier.

users
public.users

Search rows

userid	username	city	state	likesports	liketheatre	likeconcerts	likejazz
3	IFT66TXU	High Point	ME	true	false		false
8	AZG78YIP	Guayama	AK			true	true
12	FK28WAS	Kona	OH			false	
13	QTF33MCG	Bossier City	QC		true		
17	WWZ18EOX	Mobile	ON	false	false		
21	JEO48YJJ	Charlotte	NM		false	false	
26	XBG73AGE	Roswell	WI	false	true		
30	HMF84NKS	Effingham	NU		true		
31	DCF82GRJ	Nenana	NU			false	
41	KJZ19UUX	Methuen	PE		true	true	true

This is the Users table preview.

dev

Select schema info
To view tables, select a schema.

public

Filter tables

date
sales
users

```

1 SELECT sum(qtysold)
2 FROM sales, date
3 WHERE sales.dateid = date.dateid
4 AND caldate = '2008-01-05';
5

```

Run Save Schedule Clear

Send feedback

Query results Table details

Query 14085

Execution Data Visualize

Completed, started on July 13, 2024 at 15:02:07
ELAPSED TIME: 00 m 02 s

Rows returned (1)

Export

Search rows

sum
840

The above query will be used to check how much quantity has been sold and it returns as sum 840.

aws Services Search [Option+S] N. Virginia voclabs/user3314895=Jeevan_kumar_Banoth @ 7422-0413-2534

Resources info

Select database info
To view schemas, select a database.
dev

Select schema info
To view tables, select a schema.
public

Filter tables

date sales users

Query 1 Query 2 Query 3 Query 4

```
1 SELECT username, total_quantity
2 FROM
3 (SELECT buyerid, sum(qtysold) total_quantity
4 FROM sales
5 GROUP BY buyerid
6 ORDER BY total_quantity desc limit 10) Q, users
7 WHERE Q.buyerid = userid
8 ORDER BY Q.total_quantity desc;
```

Run Save Schedule Clear Send feedback

Query results Table details

Query 14100 Execution Data Visualize

Completed, started on July 13, 2024 at 15:03:10
ELAPSED TIME: 00 m 02 s

Rows returned (20) Export

This query will check for the quantity sold but it returns only 10 rows as below.

aws Services Search [Option+S] N. Virginia voclabs/user3314895=Jeevan_kumar_Banoth @ 7422-0413-2534

Query 14100 Execution Data Visualize

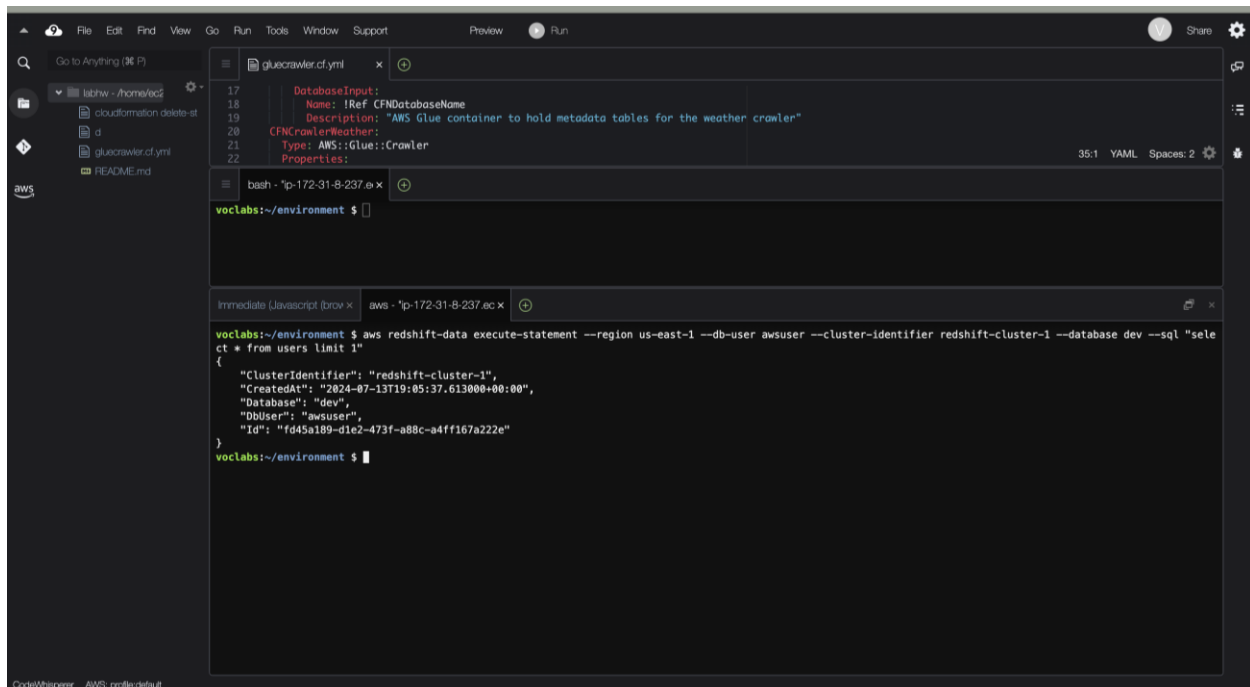
Completed, started on July 13, 2024 at 15:03:10
ELAPSED TIME: 00 m 02 s

Rows returned (20) Export

Search rows

username	total_quantity
CNF70VPH	134
CNF70VPH	134
EDB46JXK	128
KTV94TWB	128
EDB46JXK	128
KTV94TWB	128
CBC51API	126
CBC51API	126
XJN46RCL	120
HZP985GB	120

Task 6: Running Queries from the AWS CLI



The screenshot shows the AWS Cloud9 IDE interface. The top menu bar includes File, Edit, Find, View, Go, Run, Tools, Window, Support, Preview, and Run. The left sidebar shows a file explorer with a tree view containing 'cloudformation', 'gluecrawler.d.yml', and 'README.md'. The main editor area displays a YAML file named 'gluecrawler.d.yml' with the following content:

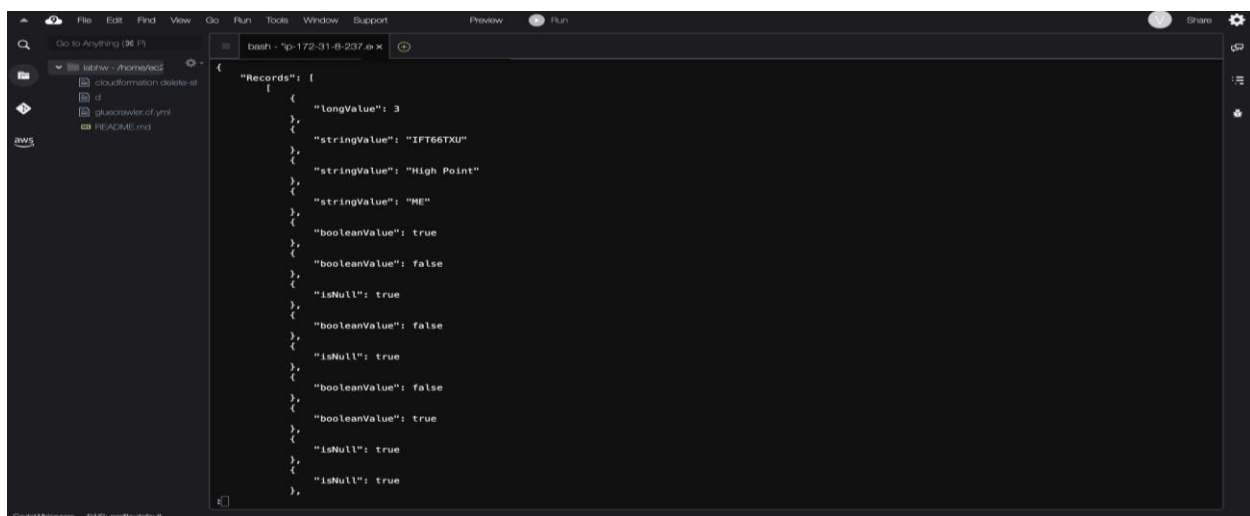
```
17 DatabaseInput:
18   Name: !Ref CFNDatabaseName
19   Description: "AWS Glue container to hold metadata tables for the weather crawler"
20   CFNCrawler@weather:
21     Type: AWS::Glue::Crawler
22     Properties:
```

Below the editor, a terminal window is open with the prompt 'voclabs:~/environment \$'. The terminal shows the command 'aws redshift-data execute-statement --region us-east-1 --db-user awsuser --cluster-identifier redshift-cluster-1 --database dev --sql "select * from users limit 1"' and its output:

```
{
  "ClusterIdentifier": "redshift-cluster-1",
  "CreatedAt": "2024-07-13T19:05:37.613000+00:00",
  "Database": "dev",
  "DbUser": "awsuser",
  "Id": "fd45a189-d1e2-473f-a88c-a4ff167a222e"
}
```

First, we will open Cloud9 and from that we will open the IDE of cloud9 as above. In the terminal we must run the command which is given in the manual, and it will display user ID and database name.

We need to copy the ID which is in the last row, and we will use that in the next command.



The screenshot shows the AWS Cloud9 IDE interface. The top menu bar includes File, Edit, Find, View, Go, Run, Tools, Window, Support, Preview, and Run. The left sidebar shows a file explorer with a tree view containing 'cloudformation', 'gluecrawler.d.yml', and 'README.md'. The main editor area displays a terminal window with the prompt 'voclabs:~/environment \$'. The terminal shows the command 'aws redshift-data execute-statement --region us-east-1 --db-user awsuser --cluster-identifier redshift-cluster-1 --database dev --sql "select * from users limit 1"' and its output:

```
{
  "Records": [
    {
      "longValue": 3
    },
    {
      "stringValue": "IFT66TXU"
    },
    {
      "stringValue": "High Point"
    },
    {
      "stringValue": "ME"
    },
    {
      "booleanValue": true
    },
    {
      "booleanValue": false
    },
    {
      "isNull": true
    },
    {
      "booleanValue": false
    },
    {
      "isNull": true
    },
    {
      "booleanValue": false
    },
    {
      "booleanValue": true
    },
    {
      "isNull": true
    },
    {
      "isNull": true
    }
  ]
}
```

In the place of Query Id, we must replace that copied one and run the command, It will going to display the above result.

Conclusion:

During this series of exercises, we worked on setting up and managing an Amazon Redshift cluster, importing data from Amazon S3, executing SQL queries, and understanding the needed IAM policies required for access control.

The main problem we encountered at first was moving data from an S3 bucket to our Redshift cluster. It was caused by an incorrect IAM role ARN at first error.

In another turn of events, we encountered yet another error where by the S3 bucket was in an other region of AWS from that of the cluster (Redshift). By including the correct region in our COPY command, we could then ascertain the appropriate regional configuration for our resources.

Once we had successfully copied the data to the Redshift Cluster, we then proceeded to interrogate this data. Before anything else we ran a straight scan of the Sales table so that we could ascertain if there were any disparities whatsoever.

This initial step was very important in confirming that indeed the importation process had been done as anticipated on the dataset.

After that, we made a command that gave us the sum or aggregate number of items sold in one day, indicating the extent of our interaction with it.

Additionally, a more elaborate command was used to fetch names of those who bought most products in amazon Sortable Track Dataset. Further we ran a more complex query to identify the top 10 buyers.

After successfully retrieving data, we investigated how to run queries through the AWS CLI. This task required programmatically executing SQL commands on the Redshift cluster via AWS Cloud9. We demonstrated the method of executing database queries outside the AWS Management Console by utilizing AWS CLI and highlighted the flexibility and automation features provided by AWS services."

As we have gone through these tasks, we have been reminded of how important it is to have detailed setups and appropriate employment of IAM roles and policies so as to guarantee safe and smooth entry into the Redshift cluster therefore facilitating its operationality in terms of handling data as well as querying processes.